

Akshaykumar Nayee

Software Engineer

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Summary

Software Engineer with 4 + years of experience building AI-powered applications, scalable backend systems, and modern web platforms. Experienced in Generative AI, LLM applications, Retrieval-Augmented Generation (RAG), cloud-native architectures, and full-stack development. Proven track record delivering production-grade solutions using Python, FastAPI, React, Angular, AWS, Docker, Kubernetes, and modern AI frameworks. Strong background in AI research, enterprise software engineering, and distributed systems.

Skills

Programming & Scripting Languages: Python (Advanced), TypeScript, JavaScript (ES6+), SQL, Bash

Frontend Engineering: React.js, Next.js 15, Angular, Vue.js, Redux, Tailwind CSS, Bootstrap

Backend Engineering: FastAPI, Django REST Framework, Flask, Node.js, gRPC, GraphQL

Databases & Storage: PostgreSQL, MongoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (Lambda, EC2, S3, CloudFormation), GCP, Azure

Docker, Kubernetes, Terraform: CI/CD (Jenkins, GitHub Actions, ArgoCD)

Data & ML Engineering: Apache Airflow, Spark, PySpark, Delta Lake, Pandas, NumPy, TensorFlow, Scikit-learn, Hugging Face, MLflow

AI / LLMs: OpenAI GPT-4, LLaMA 3.1, Claude Sonnet 4, Gemini 2.0 Flash, RAG Pipelines, Vector Databases,

Architecture & Design: Microservices, Event-Driven Architecture, RESTful & GraphQL APIs, Domain-Driven Design (DDD)

Certifications: [AWS Certified Cloud Practitioner](#) | [Lakehouse Architecture on Databricks](#) | [Generative AI](#)

Experience

Plaid, Remote, USA | Software Engineer

June 2025 – Current

- Designed and implemented microservices-based backend systems using Python, FastAPI, PostgreSQL, and AWS, improving service scalability by 40% while maintaining 99.9% platform availability.
- Built AI-powered financial recommendation and analytics engines leveraging LLMs, predictive modeling, and behavioral insights, increasing recommendation relevance by 30%.
- Developed event-driven data processing pipelines handling over 1M+ financial transactions monthly with sub-second response times.
- Optimized API performance, reducing average response latency by 45% and improving overall system throughput by 35%.
- Implemented CI/CD pipelines using GitHub Actions, Docker, Kubernetes, and AWS cloud services, reducing deployment time by 60%.
- Enhanced system observability through centralized logging, monitoring, and alerting frameworks, reducing production incident resolution times by 50%.

Gannon University, PA | Graduate Research Assistant

Jan 2024 – May 2025

- Led development of AI-powered analytics platforms for smart energy management and IoT monitoring systems, processing over 100,000 sensor events daily.
- Designed machine learning models for predictive maintenance, anomaly detection, and fault classification, improving detection accuracy by 35%.
- Built real-time dashboards and monitoring platforms using Angular, Python, FastAPI, and cloud-native architectures, reducing operator response times by 40%.
- Researched and implemented advanced AI techniques for intelligent automation, predictive analytics, and system optimization.
- Improved platform reliability to 99.5% uptime through automated monitoring, alerting, and fault recovery mechanisms.

Gannon University, PA | Student Research Assistant

Sept 2023 – Jan 2024

- Developed software tools and automation workflows supporting university research initiative and assisted in building data collection, processing, and visualization systems.
- Worked with Python, Fast API SQL, and cloud technologies to support research projects.

Rishabh Software, India | Associate Software Engineer

Sep 2021 – Jul 2023

- Developed and maintained full-stack web applications using React, Angular, Django REST Framework, and PostgreSQL across fintech and healthcare projects.
- Designed and implemented RESTful and GraphQL APIs, focusing on clean contracts, scalability, and efficient data access patterns.
- Implemented search and caching layers using Elasticsearch and Redis to improve application responsiveness.
- Created batch and scheduled data workflows using PySpark and Airflow to process large volumes of structured data reliably.

Web brains Technologies, India | Full Stack Engineer

Aug 2020 – Aug 2021

- Developed and maintained full-stack web applications using Angular, Node.js, JavaScript, PostgreSQL, and RESTful APIs.
- Built responsive user interfaces and reusable UI components, improving user experience and reducing development effort across projects.
- Improved application performance through frontend optimization, caching strategies, and database query tuning, reducing page load times by 35%.

Education

Master of Computer and information Science in Software Engineering | Gannon University, PA, USA

May 2025

Bachelor of Technology in Computer Engineering | Gujarat Technological University, India

July 2020